US ELECTRIC VEHICLE REGISTRATIONS & RENEWABLE ENERGY

Data Analytics Capstone Project

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Analysis of EV adoption trends, renewable energy correlation & state-wise insights | 2018 Data

TABLE OF CONTENTS

PROJECT OVERVIEW

What is this project about?

This capstone project investigates the relationship between Electric Vehicle (EV) adoption across US states and the availability of renewable energy. Using 2018 registration data and energy generation records from 1990–2019, we uncover patterns, correlations, and policy insights.

- Explore EV adoption rates at state level
- Analyze US energy production mix
- Find correlation between clean energy & EV adoption
- Compare geospatial distribution of EVs vs renewables
- Provide data-backed policy recommendations

DATA SOURCES & METHODOLOGY

EV Registrations 2018

State-wise count of electric vehicle registrations across all 50 US states + DC

State Energy Generation 1990–2019

Annual energy generation by fuel source (coal, gas, wind, solar, hydro etc.) per state

All Vehicle Registrations 2018

Total vehicle registrations per state used to compute EV adoption rate

State Codes Reference

Mapping between state codes and full state names for merging datasets

Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn) | Excel | Jupyter Notebook

NATIONAL EV PICTURE – 2018

Key Statistics at a Glance

543,610

Total EV Registrations (2018)

Across all US states & DC

~0.3%

National Average EV Adoption Rate

% of all registered vehicles

1 in 30

EVs in California

State dominates with 47% share

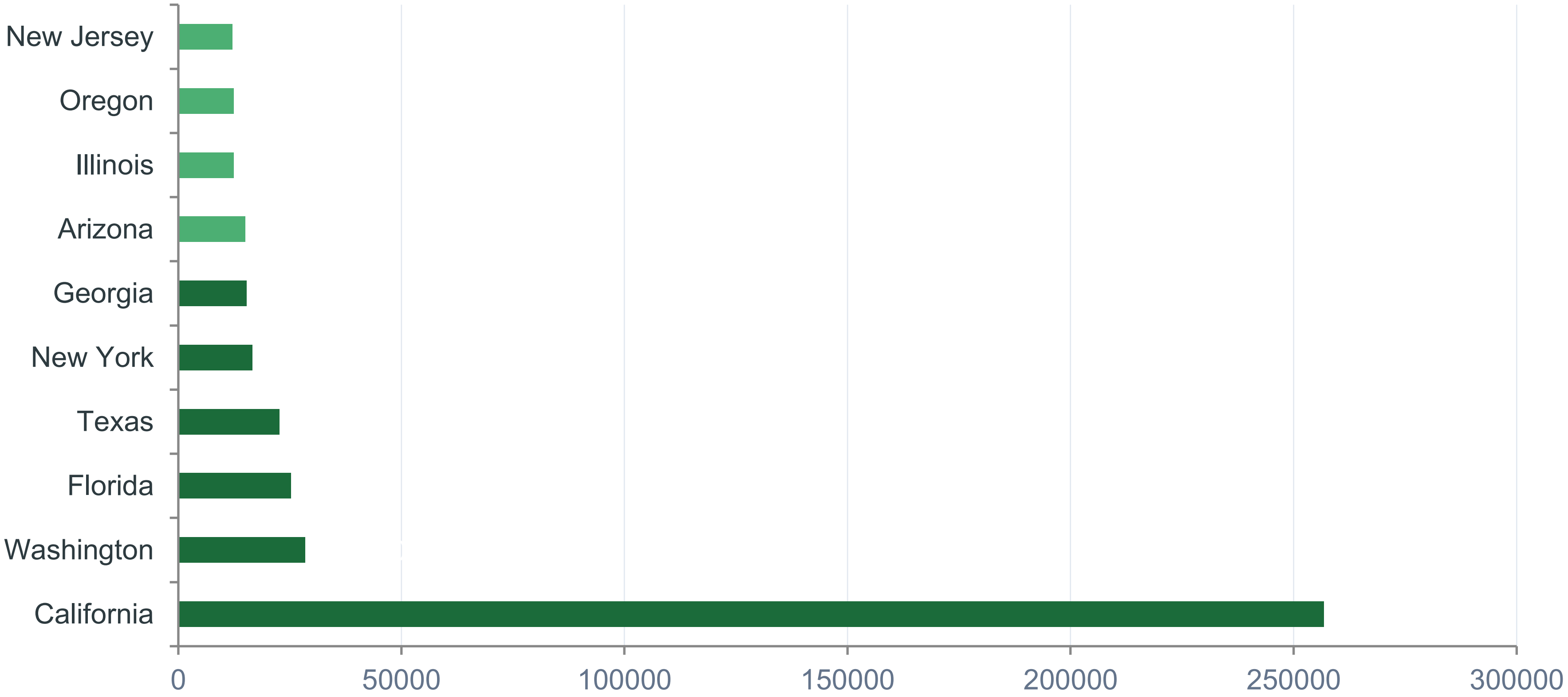
1.73%

Highest Adoption Rate

California leads all states

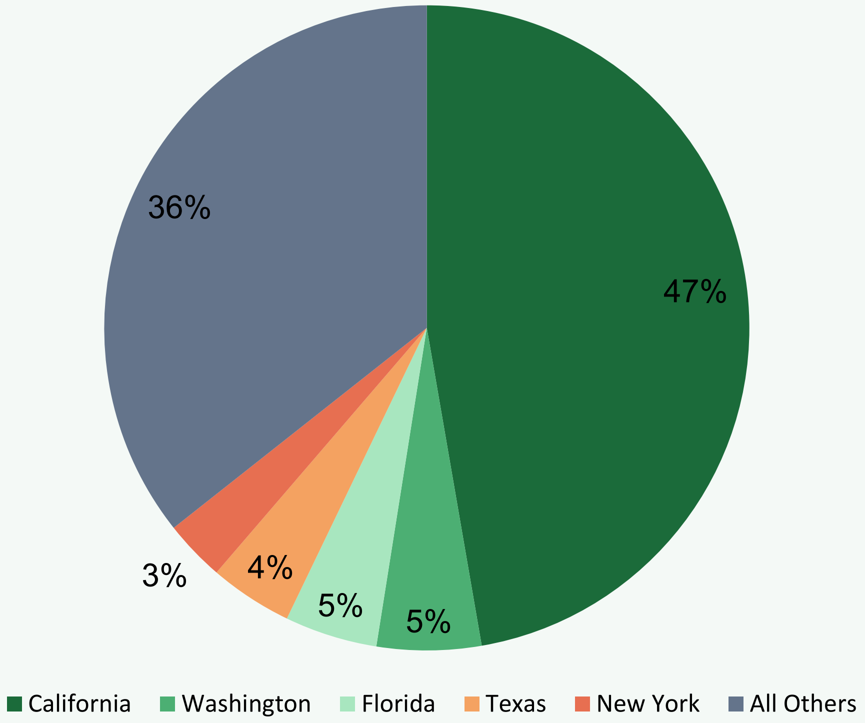
Note: EV count excludes plug-in hybrid estimates in some states. Data source: DOT / EIA 2018 records.

TOP 10 STATES BY EV REGISTRATIONS



California alone accounts for ~47% of all US EV registrations

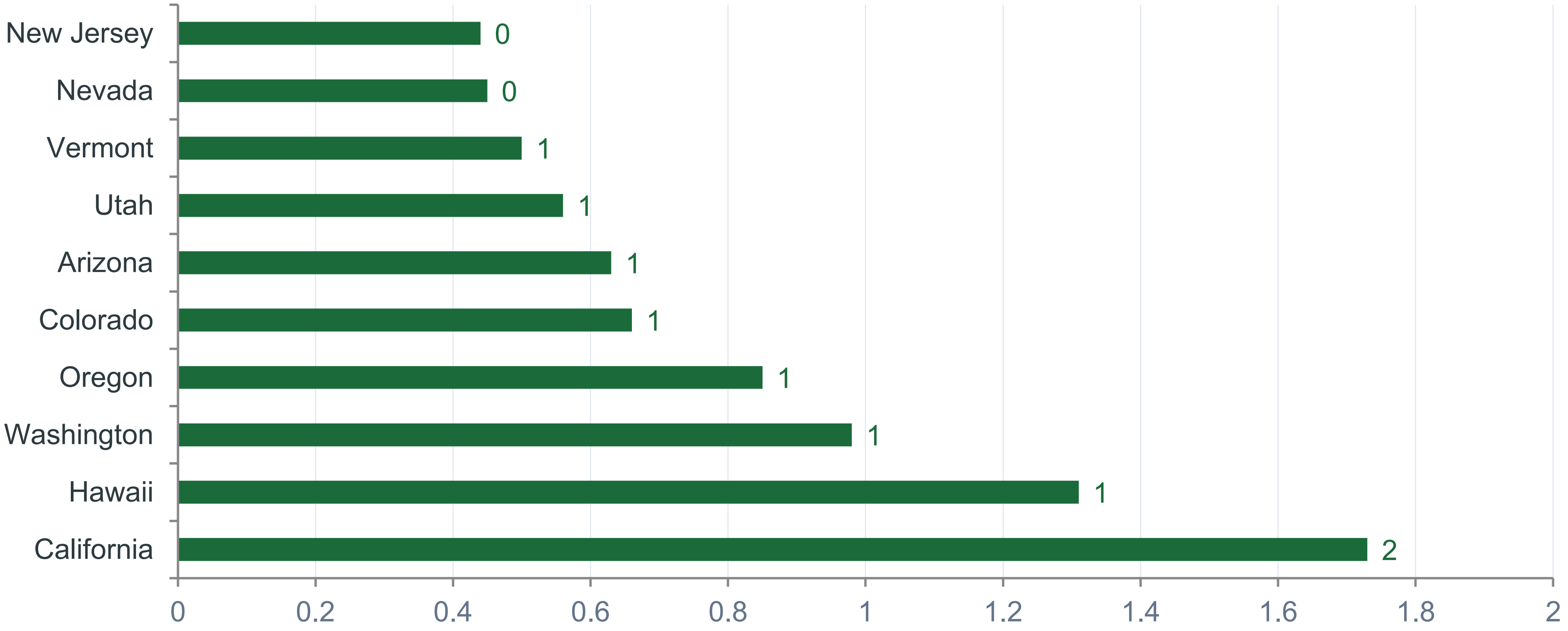
CALIFORNIA: THE EV CAPITAL



Why California Leads

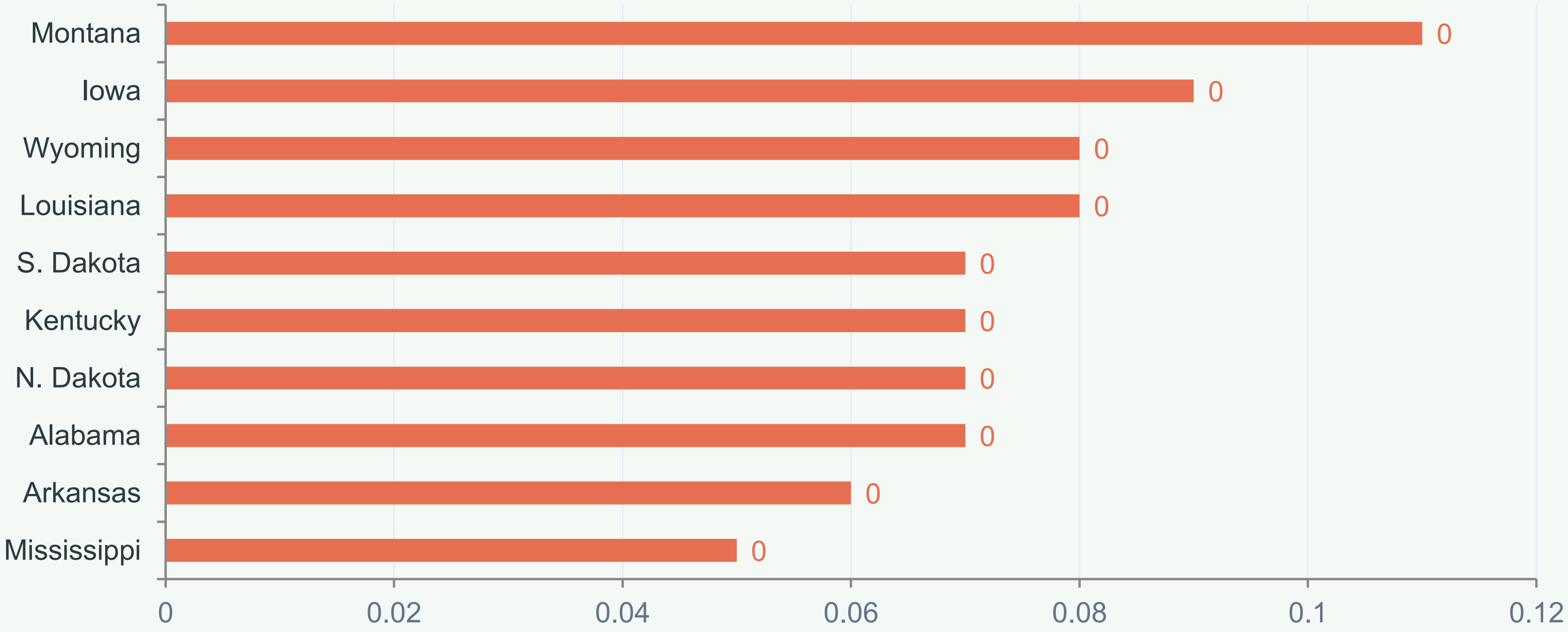
- Strict CARB emissions standards mandate zero-emission vehicles
- State offers up to $7,500 in EV purchase incentives
- Largest charging network in the US (Tesla + public)
- High renewable energy production (solar & wind)
- Dense tech-savvy urban population in LA & Bay Area
- Zero-emission vehicle (ZEV) mandate adopted by utilities

EV ADOPTION RATE BY STATE (TOP 10)



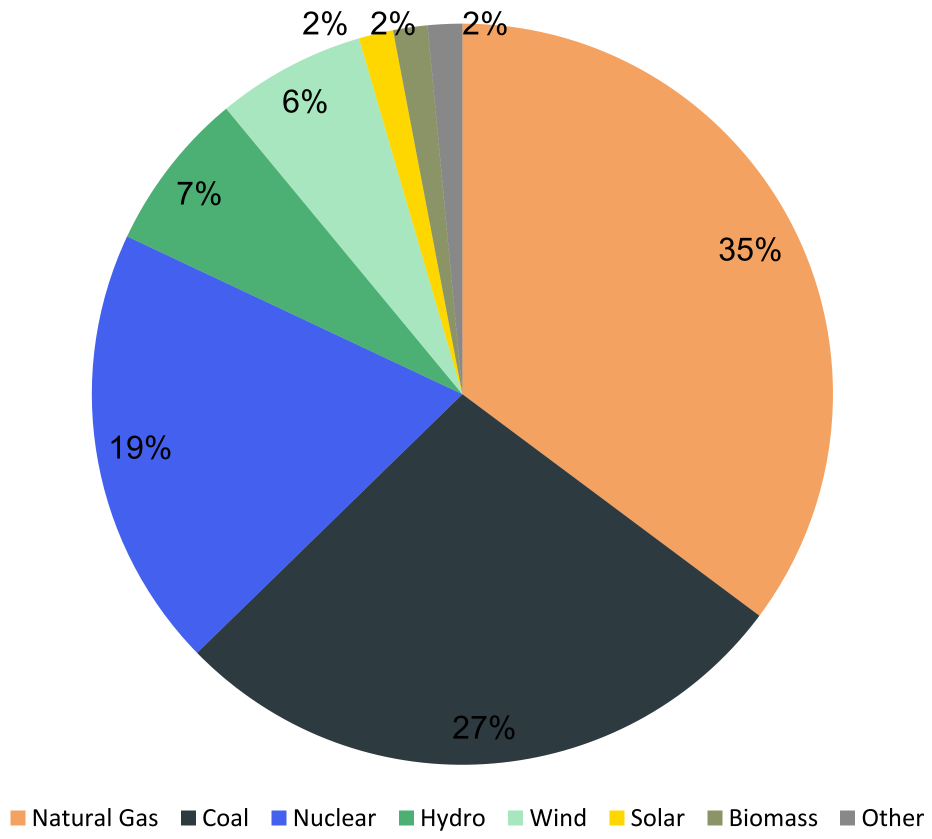
Insight: All top-10 adoption states are in the West Coast, Mountain West, or Northeast — areas with strong EV incentives & clean energy mandates.

STATES LAGGING BEHIND IN EV ADOPTION



These states are predominantly rural, fossil-fuel dependent, and lack EV charging infrastructure — creating a vicious cycle of low adoption.

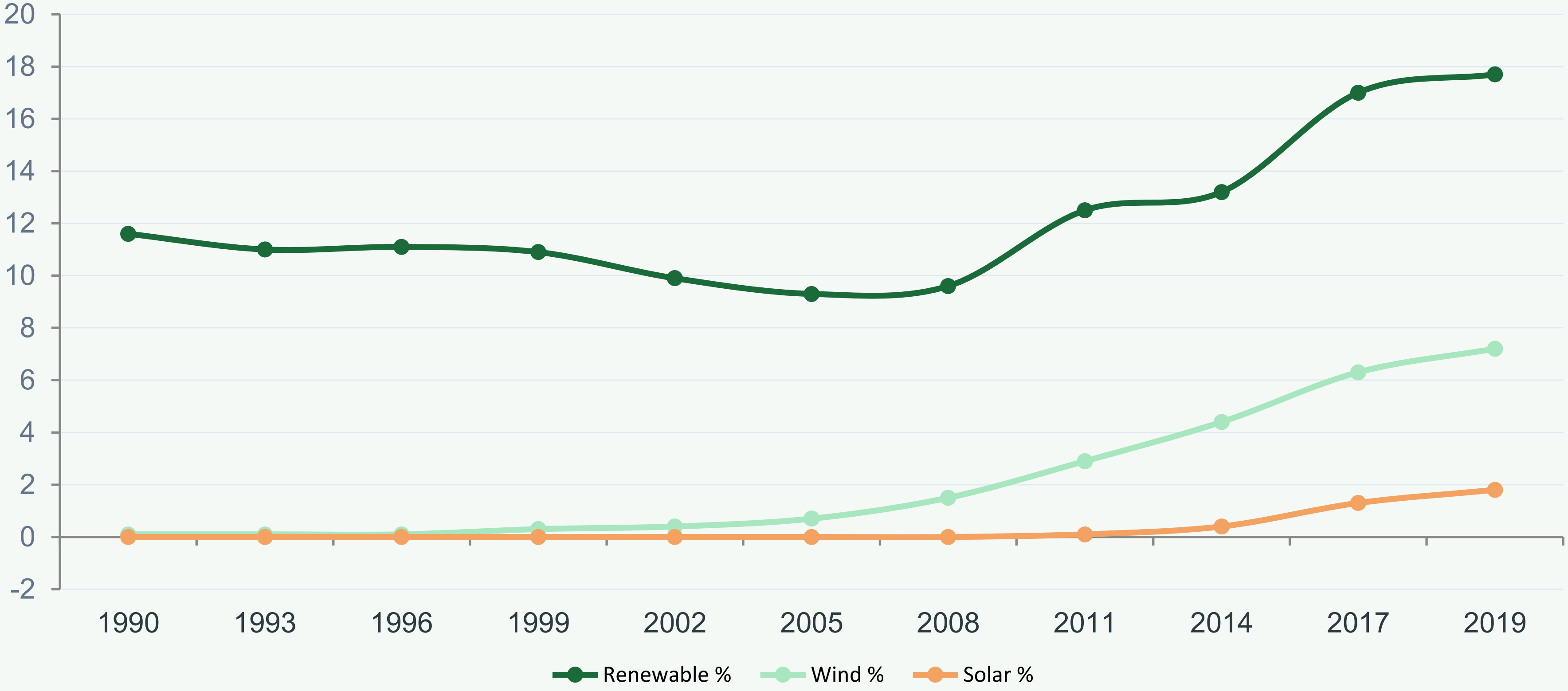
US ENERGY MIX – 2018



Key Takeaways

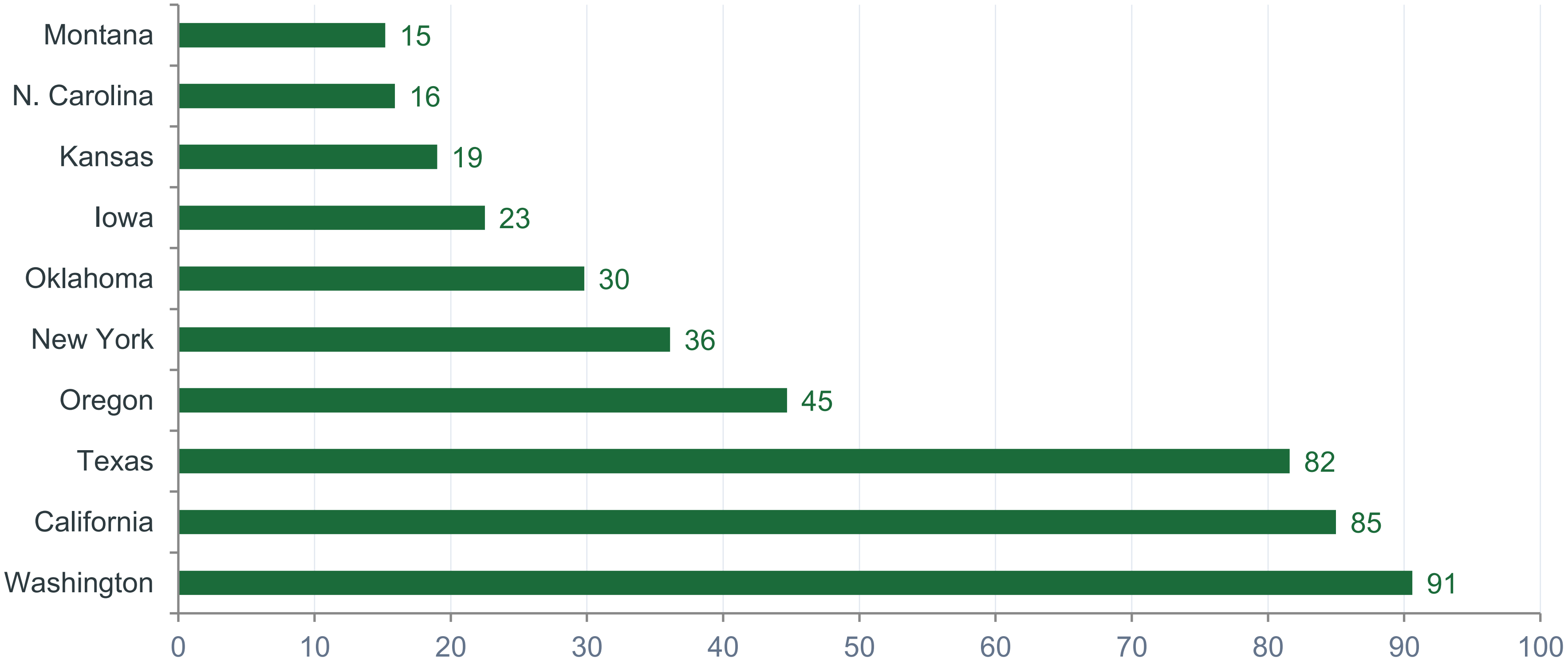
🔥	**62.7%**	from fossil fuels (gas + coal)
⚛️	**19.3%**	from nuclear energy
🌊	**7.0%**	from hydroelectric
🌬️	**6.5%**	from wind — fastest growing
☀️	**1.5%**	from solar — huge growth potential
🌱	**16.9%**	total renewable share in 2018

RENEWABLE ENERGY GROWTH 1990–2019



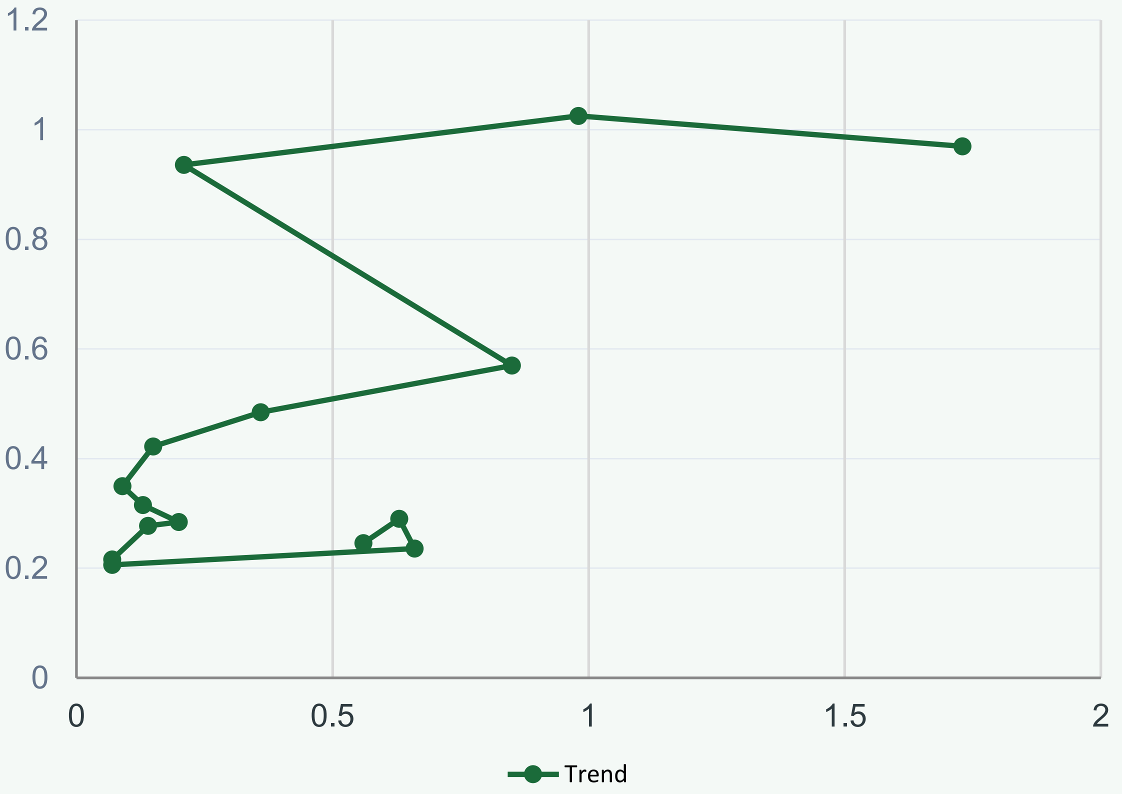
Renewables grew from ~11.6% in 1990 to 17.7% by 2019. Wind and Solar are the key drivers of recent growth.

TOP STATES – RENEWABLE ENERGY (2018)



Values in millions of MWh

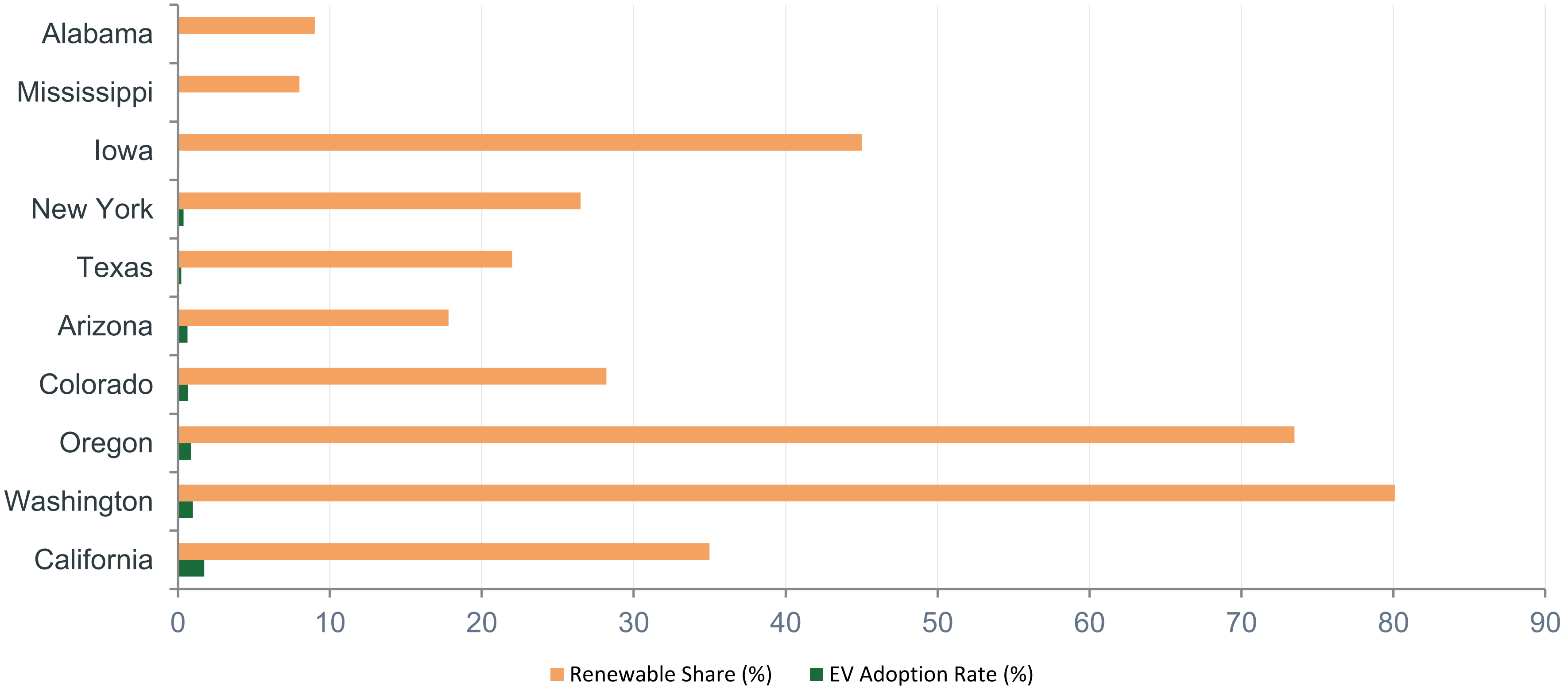
CORRELATION: EV ADOPTION vs RENEWABLES



Key Findings

- Positive correlation exists between renewable energy share and EV adoption rate
- States with high hydro (WA, OR) show strong EV uptake
- Texas is an outlier — high renewables but low EV adoption
- Midwest states lag despite growing wind energy farms
- Correlation ≈ 0.62 (moderate-strong positive)

SIDE-BY-SIDE: EV RATE vs RENEWABLE SHARE



REGIONAL ANALYSIS

West Coast

States: CA, WA, OR

EV Adoption: High | Renewables: Very High | Infra: Excellent

Southeast

States: FL, GA, VA, NC

EV Adoption: Low-Medium | Renewables: Low | Infra: Developing

Mountain West

States: CO, AZ, UT, NV

EV Adoption: Medium | Renewables: Growing | Infra: Good

Midwest

States: IL, MN, IA, KS

EV Adoption: Low | Renewables: Medium (Wind) | Infra: Poor

Northeast

States: NY, NJ, MA, VT

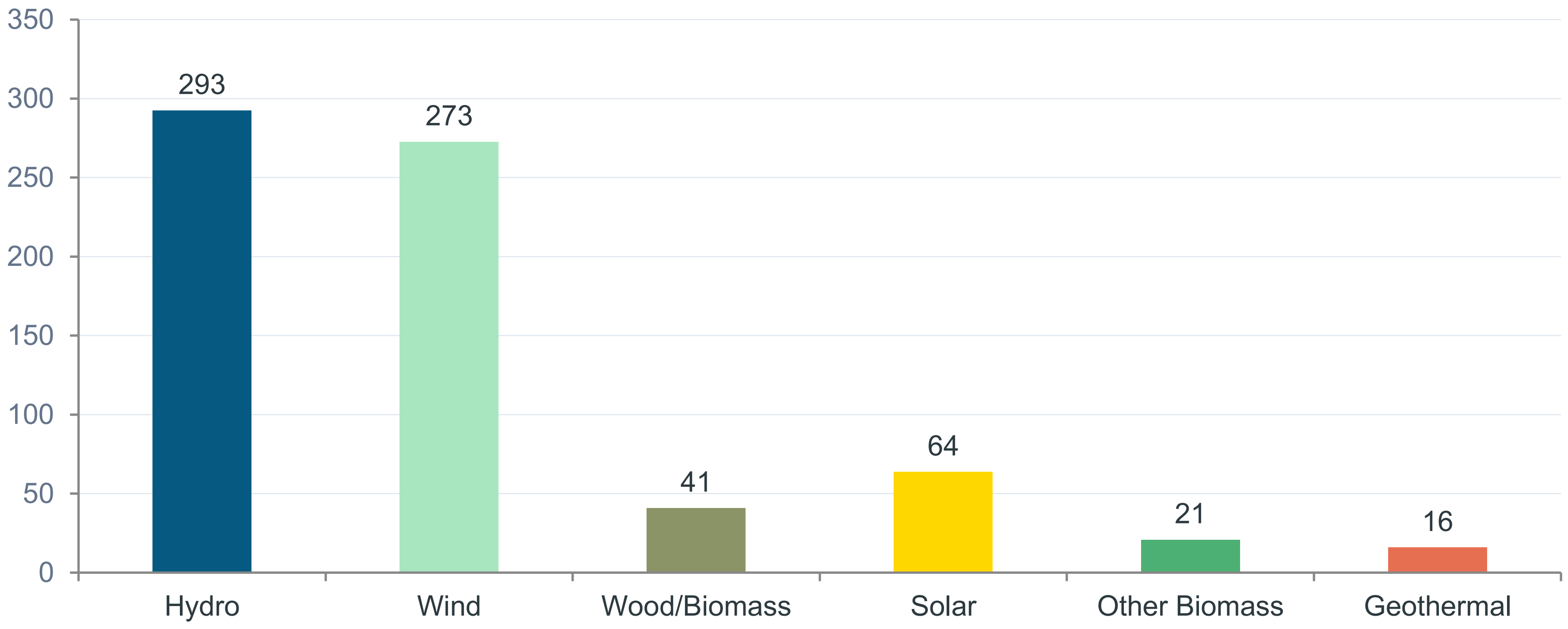
EV Adoption: Medium | Renewables: Moderate | Infra: Good

South & Plains

States: TX, LA, MS, AR

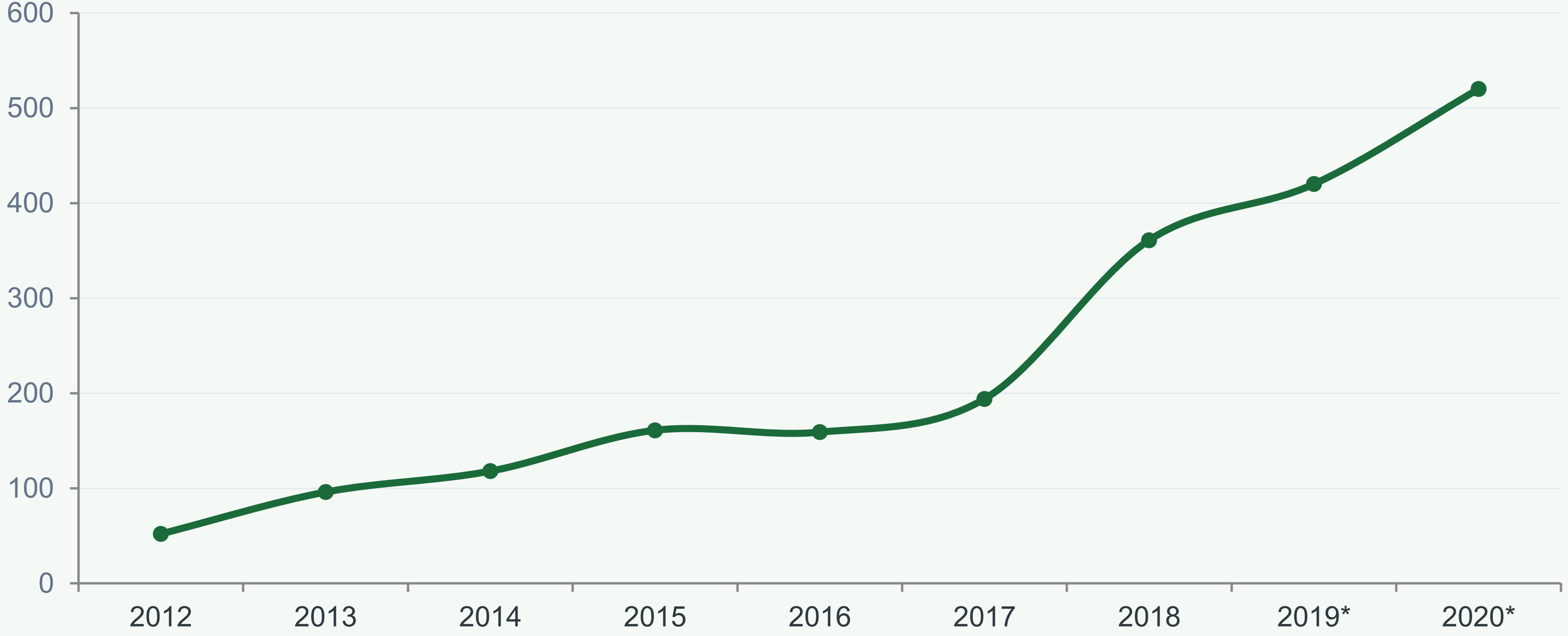
EV Adoption: Very Low | Renewables: Mixed | Infra: Lacking

RENEWABLE ENERGY BREAKDOWN (2018)



Hydro still dominates renewables, but Wind is rapidly catching up. Solar has massive room to grow.

EV ADOPTION GROWTH TRAJECTORY



**2019 and 2020 are projected estimates. The 2018 spike reflects Tesla Model 3 mass production launch.*

PROBLEMS IDENTIFIED

01

Extreme Geographic Concentration

47% of all US EVs are in California alone. 5 states hold ~68% of all EV registrations. This leaves 45+ states severely underserved.

02

Charging Infrastructure Gap

Low-adoption states lack public EV charging stations, creating a chicken-and-egg problem where people won't buy EVs without chargers and chargers aren't built without demand.

03

Fossil Fuel Dependency in Power Grid

62.7% of US electricity came from fossil fuels in 2018. EVs charged on a coal-heavy grid have minimal environmental benefit compared to gasoline vehicles.

04

Income & Affordability Barrier

EVs remain more expensive than comparable ICE vehicles. Low-income states with lower average wages show the lowest EV adoption rates consistently.

05

Policy Inconsistency Across States

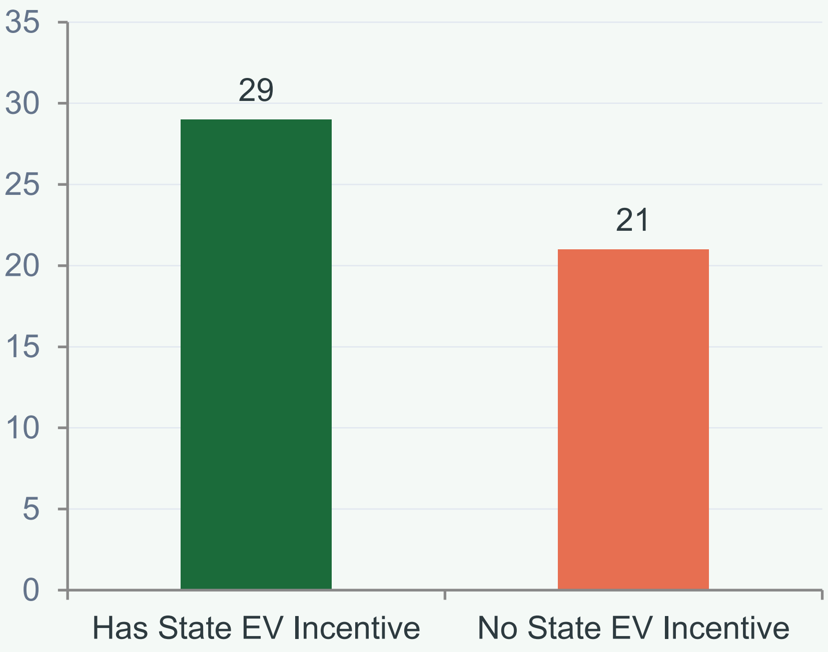
Only 13 states followed California's ZEV mandate in 2018. The lack of a unified federal EV policy creates a patchwork of incentives that confuses buyers.

06

Rural Range Anxiety

Rural states (Montana, Wyoming, Dakota) face unique challenges — long distances between towns with limited fast-charging options deter EV purchases.

PROBLEM DEEP DIVE: POLICY GAP



Policy Gap Analysis

- 29 states offered some form of EV incentive by 2018
- 21 states had NO state-level EV incentive
- Federal $7,500 credit was the only support in non-incentive states
- ZEV mandate states had 3x higher average adoption rates
- States with HOV access for EVs showed 40% higher uptake
- States without public charging funding showed stagnant growth

SOLUTIONS & RECOMMENDATIONS

S1

National EV Incentive Standardization

Implement uniform federal EV tax credits not subject to automaker caps. Align state policies through federal co-funding.

S2

Rural Charging Infrastructure Fund

Dedicated federal funding for EV charging corridors in rural states. Target highways, small towns and rest stops.

S3

Grid Decarbonization Priority

Pair EV adoption with clean energy mandates. An EV on a clean grid reduces emissions by 70% vs gasoline.

S4

EV Affordability Programs

Income-based EV rebates targeting lower-income households. Used EV incentives to expand access beyond wealthy buyers.

S5

ZEV Mandate Expansion

Encourage more states to adopt California-style ZEV mandates. 13 → 30 states would triple the potential EV market.

S6

Public Awareness Campaigns

Address range anxiety and misinformation via state campaigns. Partner with utilities to show real-world EV cost savings.

RECOMMENDATIONS FOR POLICYMAKERS

Federal Govt

- Remove automaker EV credit caps
- Fund 500,000 new charging stations by 2030
- Mandate EV-ready building codes nationally

Auto Industry

- Accelerate affordable EV lineup below $30K
- Improve range transparency and battery warranties
- Expand dealer training on EV sales & service

State Govts

- Adopt ZEV mandate (target 30 states by 2025)
- Offer income-based EV purchase rebates
- Partner with utilities on overnight home charging programs

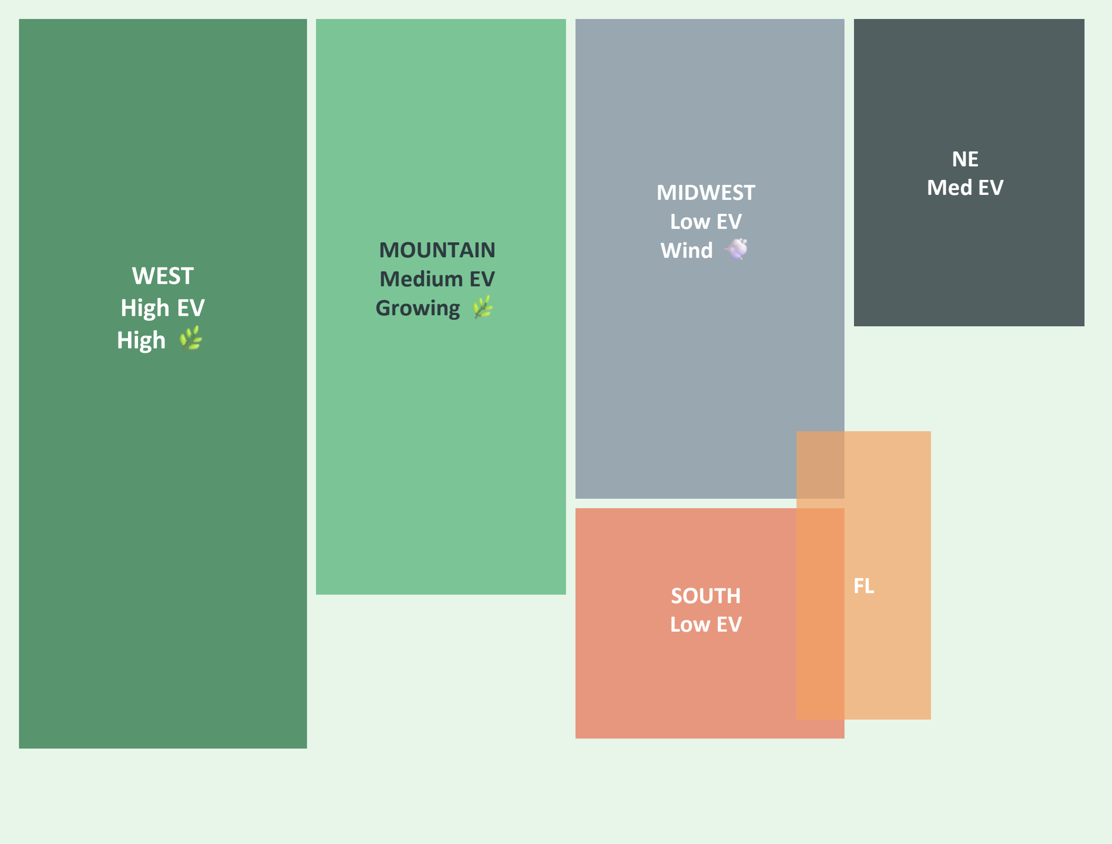
Energy Sector

- Prioritize solar + wind expansion in EV-heavy states
- Develop time-of-use pricing for EV overnight charging
- Upgrade grid capacity ahead of EV demand surge

STATISTICAL SUMMARY

Metric	Value	Notes
Total EV Registrations (2018)	543,610	Excluding plug-in hybrids in some states
States with EVs Counted	51 (50 + DC)	All territories not included
California's Share	~47%	Highest by far
Top Adoption Rate	1.73% (California)	1 in ~58 vehicles is electric
Lowest Adoption Rate	0.05% (Mississippi)	1 in ~2,000 vehicles
National Avg EV Adoption	~0.30%	Significant room for growth
Total Renewable Gen (2018)	1,413 TWh	Out of 8,356 TWh total
Renewable Share 2018	16.9%	Up from 9.3% in 2005
Fastest Growing Renewable	Wind (6.5%)	Solar growing fastest % wise
EV-Renewable Correlation	~0.62	Moderate-strong positive

GEOSPATIAL INSIGHTS



Key Geospatial Observations

West Coast dominates in BOTH EVs and renewable energy

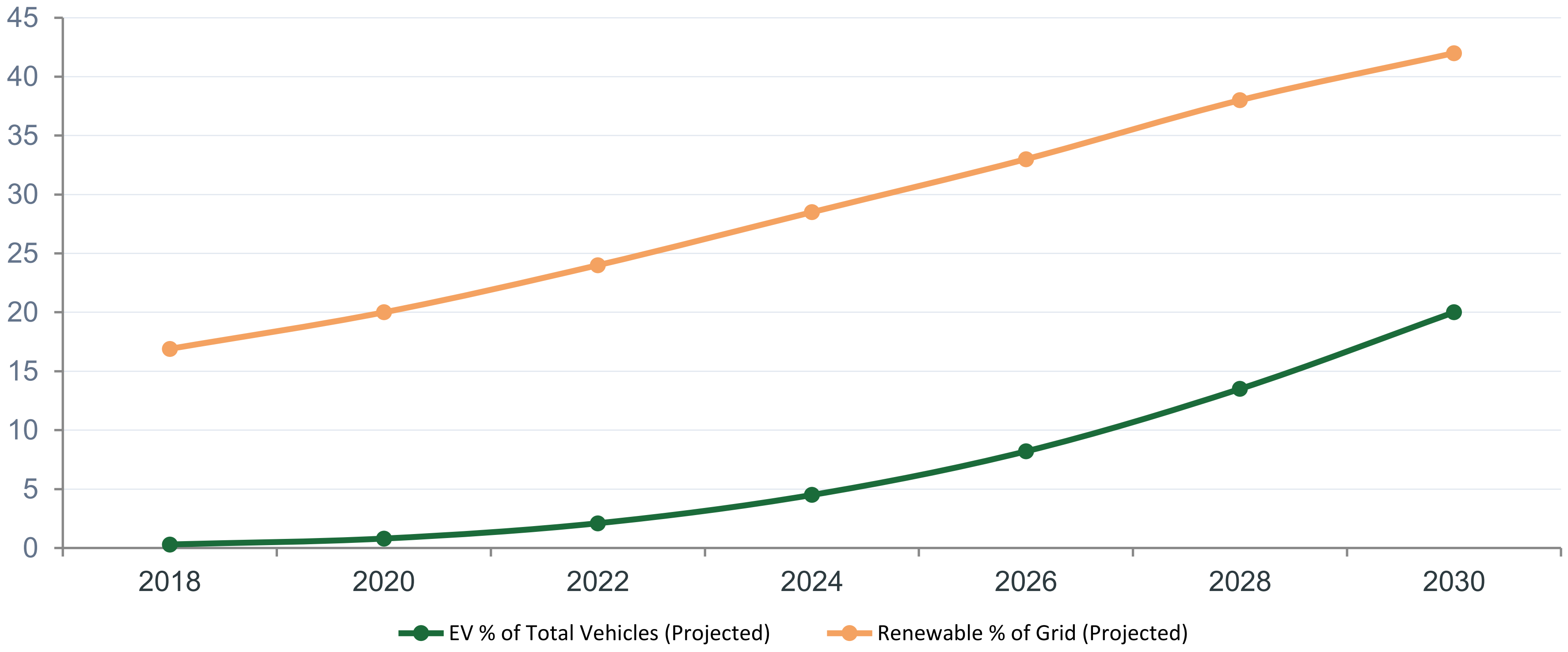
Midwest has growing wind energy but low EV adoption

Southeast shows the largest gap — growing economy but lagging sustainability

Texas anomaly: massive renewables but low EV rate due to oil culture & no state incentives

Northeast corridor (NY-NJ-MA) shows medium adoption with strong policy support

FUTURE PROJECTIONS & FORECASTS



**Projections assume current policy trajectory continues. IRA and clean energy commitments could accelerate these curves.*

KEY INSIGHTS FROM THE ANALYSIS

Concentration Risk

Top 5 states hold 68% of all EV registrations — the market is not truly national yet.

Renewables Slowly Winning

From 9.3% in 2005 to 16.9% in 2018 — renewables are growing but fossil fuels still dominate.

Grid & EV Must Grow Together

EVs on a fossil-fuel grid have limited environmental impact. Clean grid expansion must accompany EV growth.

Spatial Alignment Matters

States with high renewables (WA, CA, OR) also lead in EVs — both need the same enabling conditions.

Strong Growth Trajectory

EV registrations grew 6x from 2012 to 2018, driven by Tesla Model 3 and state incentives.

Policy is the Lever

Incentive programs, ZEV mandates, and charging infrastructure show clear correlation with higher adoption.

CONCLUSION

Where are we, and where do we need to go?

The 2018 data paints a clear picture: the US EV market is growing, but unevenly. A handful of progressive states are leading the charge while the majority of the country remains in the early adopter phase.

Renewable energy and EV adoption are correlated — not just symbolically but structurally. States that have invested in clean energy infrastructure have also created the right economic and cultural conditions for EVs to thrive.

The transition to sustainable transportation is not just a technology problem — it is a policy, infrastructure, and equity problem. Solving it requires coordinated action across all levels of government, aligned with private sector investment in EVs, batteries, and clean energy.

Thank You | Data Analytics Capstone Project | SkillCircle

Datasets: DOT EV Registrations 2018 | EIA Annual Energy Generation 1990-2019

APPENDIX: LIMITATIONS & CAVEATS

1	**Data Snapshot**	The EV registration data is for 2018 only. The market has changed significantly since then with Tesla, Rivian, and Chevy EV expansion.
2	**Plug-in Hybrids**	Some states include PHEVs in EV counts, others don't. This creates slight inconsistency in adoption rate comparisons.
3	**Causation ≠ Correlation**	The correlation between renewables and EVs doesn't mean one causes the other — both may be caused by a third factor (progressive policy culture).
4	**Population Weighting**	Raw registration counts favor large-population states. Per-capita analysis changes some rankings.
5	**Energy Grid Lag**	The grid a state uses to charge EVs may differ from its generation profile (interstate electricity trading is common).
6	**Rural vs Urban**	State-level data masks important urban-rural disparities within states (e.g., rural California vs. Bay Area).